

BERT Search Hyperparameters

Default settings for the encoder structured-pruning search.

Objective. This appendix records the main hyperparameters used by the BERT structured-pruning search. It includes the core algorithm settings, archive limits, elitism rules, stagnation behaviour, EXP3-IX bandit update, and the initial operator-prior overrides attached to the discovered island types. Migration settings are omitted because they are not part of the reported BERT configuration.

Core Search

HP	Value
pop_size	120
num_gens	80
elite_size	15
num_islands	1

Archives

HP	Value
hof_capacity	50
hof_max_per_macro	3; env: PRUNE_HOF_MAX_PER_MACRO
pareto_size	100

Elites

HP	Value
elite_size	15, passed in
Per-island elites	$\max(2, \text{elite_size}/\text{num_islands})$
Micro/macro split	$\text{macro_frac} = \min(0.80, 0.35 + 0.03 \text{isl.stag})$
Crossover rate	0.25 micro, 0.65 macro, 0.40 default

Stagnation

HP	Value
Restart threshold	$\text{isl.stag} \geq 20$
Extra full evals	24; env: PRUNE_STAG_EXTRA_FULL_EVALS

EXP3-IX Bandit

HP	Value
gamma	0.07
Initial weights	$w[o] = 1.0$ for all operators
Probability	$(1 - \gamma) w[o] / \sum_j w[j] + \gamma/K$
Update	$w[o] \leftarrow w[o] \exp\left(\gamma \frac{r}{p + \gamma} \frac{1}{K}\right)$
Overflow clip	normalise when $\max_o w[o] > 10^6$

Operator Priors

Initial overrides. The following values override the default $w[o] = 1.0$ only for the corresponding island type. Operators not listed keep weight 1.0.

Operator	Weight
budget_neutral_swap_neurons_for_head	3.0
budget_neutral_swap_head	2.0
head_neuron_trade	1.5
layer_rewire	1.0
jitter	1.0
equilibrium_mutate	0.8

Operator	Weight
budget_neutral_swap_neurons_for_head	2.0
budget_neutral_swap_head	1.8
budget_neutral_swap_neurons	1.4
equilibrium_mutate	1.4
head_neuron_trade	1.2
jitter	1.0

Operator	Weight
budget_neutral_swap_head	1.8
budget_neutral_swap_neurons	1.8
equilibrium_mutate	1.6
group_reallocate	1.2
macro_budget_shift	1.2
jitter	1.0

Operator	Weight
budget_neutral_swap_head_for_neurons	3.0
budget_neutral_swap_neurons	2.2
neuron_rebalance	1.6
group_reallocate	1.2
jitter	1.0

Operator	Weight
drop_layer	3.0
drop_multi	2.5
layer_rewire	2.0
macro_budget_shift	1.8
budget_neutral_swap_head	1.2
budget_neutral_swap_neurons_for_head	1.2
jitter	1.0